Task 3 — Instagram Engagement Analysis

Alfido Tech Data Science Internship | Project report

Task 3: Instagram Engagement Analysis

Dataset and methodology

• Source: locally available CSV in `data/raw/`.

• Findings describe this dataset only; they are not automatically generalizable.

Findings

• Rows: 29,999; columns: 15.

• Duplicate rows: 0.

• Cells missing values: 0.

• engagement_rate: mean 14.42; median 9.19; valid rows 29,999.

• likes: mean 99,912.66; median 99,580.00; valid rows 29,999.

• comments: mean 5,017.78; median 5,042.00; valid rows 29,999.

• shares: mean 2,502.91; median 2,498.00; valid rows 29,999.

• saves: mean 7,490.12; median 7,483.00; valid rows 29,999.

• reach: mean 996,561.60; median 992,181.00; valid rows 29,999.

• impressions: mean 1,246,668.83; median 1,241,026.00; valid rows 29,999.

• followers_gained: mean 502.15; median 501.00; valid rows 29,999.

• Highest average engagement rate by content category: Beauty (15.664).

• Highest average engagement rate by media type: Video (14.620).

Outputs

Charts and supporting CSV files are saved in outputs/.

Recommendations and next steps

• Compare engagement across content categories and media types.

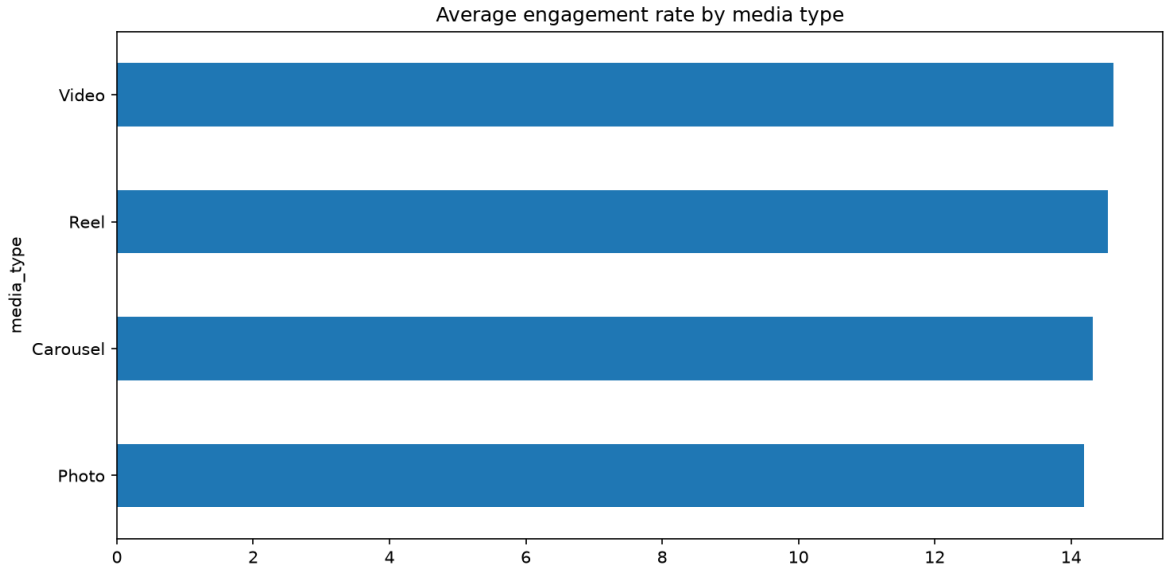
• Analyze posting days and times only when timestamps are available and valid.

• Separate likes, comments, shares and saves when those fields exist.

• Test hashtags and captions systematically; do not claim causation from simple correlations.

• Update the content calendar using verified account-specific results.

Task3 Instagram Media Engagement



Task3 Instagram Category Engagement

